

Below, we present additional experiments conducted during the rebuttal period, which address reviewers’ comments on dependence on dense rewards, additional baselines, generalization to other games, and robustness across VLM backbones. Across these settings, **Odysseus** consistently achieves stable and efficient learning and outperforms the corresponding baselines. Some experiments are still running, and we will include the full results in the revised paper.

Mario with coarser milestone rewards.

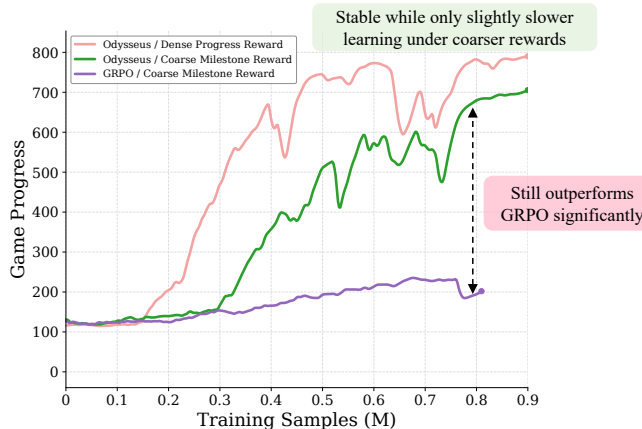


Figure 1: We re-evaluate the Mario task from Sections 4 and 5 using a **coarser milestone reward**. Instead of the dense x-coordinate progress reward, this reward only indicates whether the agent passes key pipes or milestones. **Odysseus** still achieves stable and effective training in this setting, with only slightly slower learning than under the dense progress reward, while continuing to outperform the GRPO baseline. This result suggests that **Odysseus does not fundamentally rely on fully dense progress rewards**.

Additional baselines.

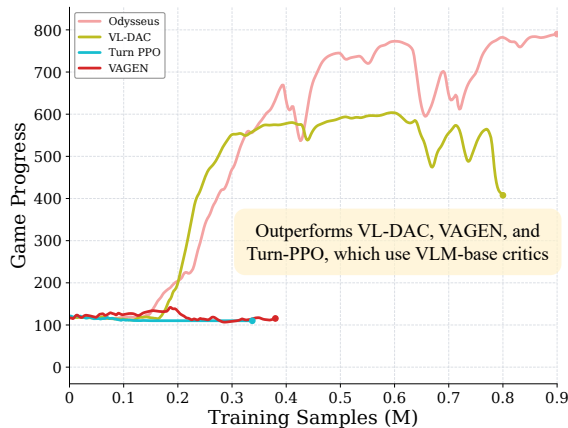
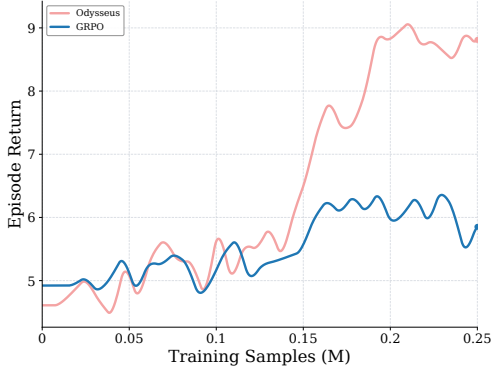
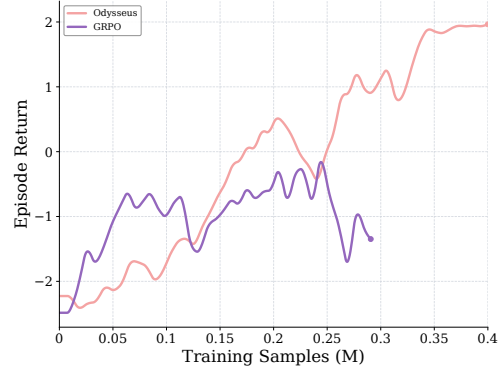


Figure 2: Following the reviewers’ suggestion, we implemented VL-DAC from [Ref 1], VAGEN from [Ref 2], and Turn-PPO from [Ref 5], as additional baselines, which use VLM-based critics. In terms of performance, **Odysseus still outperforms VL-DAC, VAGEN, and Turn-PPO**. Moreover, under the same batch-size configuration, VL-DAC, VAGEN, and Turn-PPO require roughly twice the GPU resources of **Odysseus** in our experiments. This highlights the **resource efficiency of Odysseus**, which benefits from its lightweight CNN-based critic.

Generalization beyond Mario.



(a) Procgen Coinrun with outcome rewards.



(b) Procgen Fruitbot with coarse milestone rewards.

Figure 3: We added two new games to test the generality of Odysseus beyond the Super Mario Land setting studied in the main paper: Procgen CoinRun with outcome rewards and Procgen FruitBot with coarse progress rewards. These games differ from Mario in visual appearance, game dynamics, and reward structure. Odysseus still achieves stable and efficient learning in both environments, suggesting that **the proposed training recipe is not specific to the original Mario setup.**

Different VLM backbones.

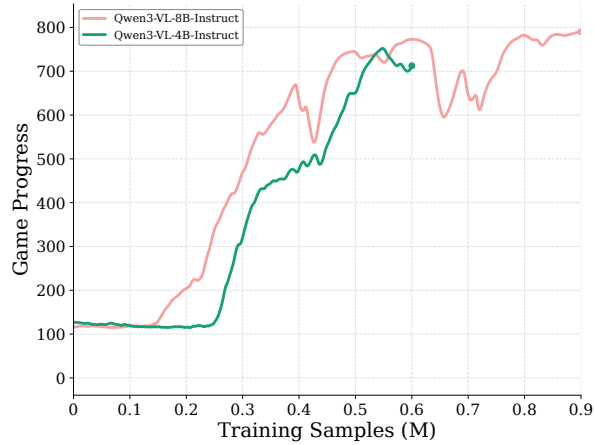


Figure 4: Experiments applying Odysseus to Qwen3-VL-4B-Instruct indicate that **the proposed training recipe is not specific to the Qwen3-VL-8B-Instruct model** mainly studied in the paper.

Ablations on handling negative advantages.

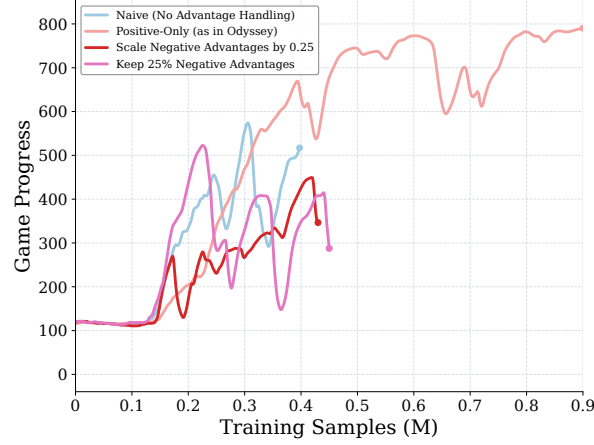


Figure 5: We performed additional ablations on how to handle negative advantages, beyond the two settings already included in the paper: positive-advantage filtering, as used in Odysseus, and the naive approach with no special handling. Specifically, we evaluate two softer alternatives: scaling negative advantages by a factor of 0.25, and retaining only the 25% of negative advantages that are closest to zero. The results show that the positive-only variant, i.e., the positive-advantage filtering used in Odysseus, remains a **simple yet effective approach** for achieving stable and efficient training.

[Ref 1] Bredis, George, et al. "Enhancing Vision-Language Model Training with Reinforcement Learning in Synthetic Worlds for Real-World Success." 2025.

[Ref. 2] Wang, Kangrui, et al. "VAGEN: Reinforcing World Model Reasoning for Multi-Turn VLM Agents." 2025.

[Ref. 5] Li, Junbo, et al. "Turn-PPO: Turn-Level Advantage Estimation with PPO for Improved Multi-Turn RL in Agentic LLMs." 2025.